Stationary-autocorrelation audit of the three-mode relaxation rate

7 September 2026

Question and verdict

We replaced the window-dependent fit of conditional expectations by stationary autocorrelations

$$C_g(t) = \langle g(X_s)g(X_{s+t})\rangle_s - \langle g\rangle^2, \qquad \lambda_{\text{eff}}(t) = \frac{d}{dt}\log C_g(t).$$

The predeclared test does **not** find a common plateau across the seven observables. It therefore does not resolve a generator-wide spectral gap. Two driven action observables have individually resolved rates near -1, but pooling them into a spectral-gap estimate would exceed the evidence.

Frozen simulation and analysis

The Euler–Maruyama map is copied from `cpp/backward/NLS_backward.cpp`: $n = 3$, $\gamma = 0.1$, $dt = 10^{-3}$, canonical endpoint baths, single-precision state, inherited polynomial trigonometry and positive-action projection. The driven case is $(T_1, T_3) = (2, 8)$ and the reference case is $(5, 5)$. Each case uses 64 explicitly seeded independent trajectories, burn-in 500, stationary measurement time 1000 per trajectory, and snapshots every 0.01. Thus the aggregate stationary time is 64,000 and the largest lag 10, exceeding the requested $200\,t_{\max}$ requirement by a factor 32.

The lag grid is the set of unique rounded indices from `geomspace(1,1000,121)`. We use the centred five-grid-point finite difference

$$\lambda_{\text{eff}}(t_i) = \frac{\log C(t_{i+2}) - \log C(t_{i-2})}{t_{i+2} - t_{i-2}}.$$

The frozen plateau requires $t \geq 1$, $C/\text{SE} \geq 3$, at least eight contiguous points spanning a factor 1.5 in time, and no more than 20% rate variation. All uncertainty intervals resample the 64 trajectories, never the overlapping time origins.

Driven result

g	$C_g(0)$	plateau	λ_R	95% CI	N_{eff}
$\cos\theta_1$	0.4794	—	—	—	—
$\sin\theta_1$	0.4679	—	—	—	—
$\cos\theta_3$	0.4872	—	—	—	—
$\cos(\theta_1 - \theta_3)$	0.4999	—	—	—	—
I_2	2.6107	$[1.12, 2.37]$	-0.9926	$[-1.0517, -0.9337]$	56,442
$I_1 + I_3$	3.1273	$[1.00, 1.58]$	-1.1088	$[-1.2013, -1.0228]$	74,921
$I_1 - I_3$	5.1962	—	—	—	—

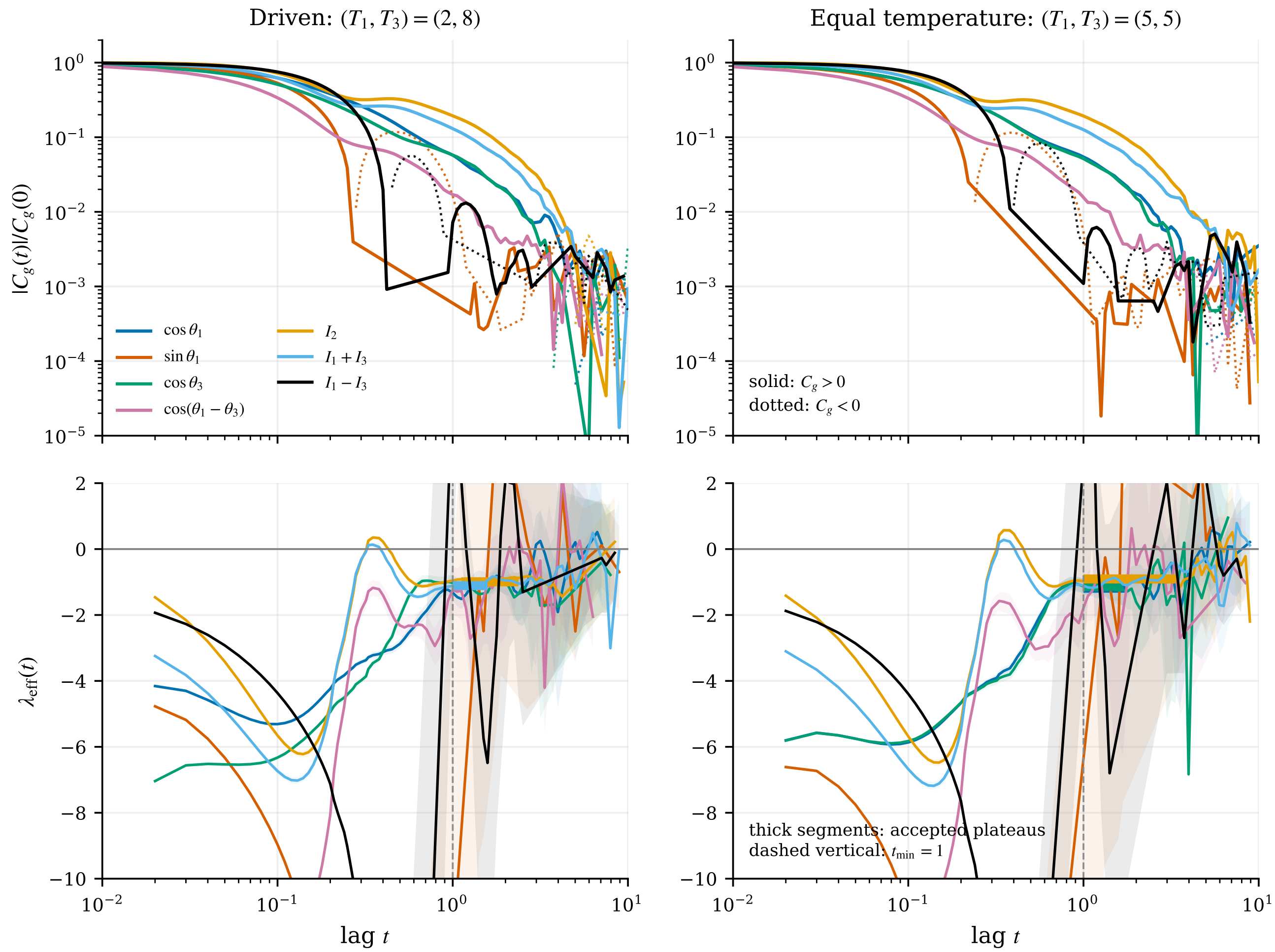

Figure 1: Top: stationary autocorrelations normalized by $C_g(0)$ on a logarithmic magnitude axis. Solid portions are positive and dotted portions negative. Bottom: local decay rates with trajectory-bootstrap 95% bands. Thick line segments mark intervals accepted by the frozen plateau rule. The rapidly growing bands show where the correlation has entered its noise floor.

Only two observables pass, below the predeclared minimum of three. Moreover, the remaining correlations either change sign or lose 3 SE support before a stable late interval emerges. The driven result is therefore *observable-specific plateaus, no common plateau.*

Equal-temperature reference

g	$C_g(0)$	plateau	λ_R	95% CI	N_{eff}
$\cos\theta_1$	0.4879	[1.00, 1.78]	−1.1878	[−1.3607, −1.0160]	122,898
$\sin\theta_1$	0.4787	—	—	—	—
$\cos\theta_3$	0.4863	[1.00, 1.68]	−1.1334	[−1.3366, −0.9465]	126,444
$\cos(\theta_1-\theta_3)$	0.4998	—	—	—	—
I_2	2.4254	[1.00, 3.35]	−0.9065	[−0.9665, −0.8442]	54,591
I_1+I_3	3.0387	—	—	—	—
I_1-I_3	4.9834	—	—	—	—

Three observables pass separately, but the estimates span -1.188 to -0.907, exceed the allowed 20% spread, and have no common 95% CI intersection. Hence the equilibrium common-plateau gate also fails.

An important correction to the proposed control is that Gibbs invariance does not make this full generator self-adjoint: its Hamiltonian Liouville component is skew-adjoint in Gibbs L^2. Consequently an equilibrium autocorrelation need not be everywhere positive or monotone. We report sign changes and monotonicity violations in the raw diagnostics, but do not misclassify them as automatic numerical artefacts.

For completeness, the raw equal-temperature diagnostics are:

g	significant rises	supported negative lags	first negative t
$\cos\theta_1$	11	0	—
$\sin\theta_1$	85	78	0.24
$\cos\theta_3$	5	0	—
$\cos(\theta_1-\theta_3)$	24	0	—
I_2	18	0	—
I_1+I_3	18	0	—
I_1-I_3	81	53	0.40

Table 1: Equal-temperature sign and monotonicity diagnostics. A “significant rise” means an adjacent-lag increase larger than the propagated standard error; a supported negative lag has $C_g < 0$ with $|C_g|/\text{SE} \geq 3$. These are descriptive diagnostics, not rejection gates for the non-reversible Hamiltonian-plus-bath generator.

Burn-in and effective support

All fourteen normalized RMS differences between the full measured trajectory and the same calculation after deleting its first quarter are below 0.004, against the frozen 0.05 gate. The integrated-autocorrelation estimate gives more than 50,000 effective samples for every accepted plateau. The failure to find a common rate is thus not a small-effective-sample failure; it is caused by observable dependence and loss of late-time signal.

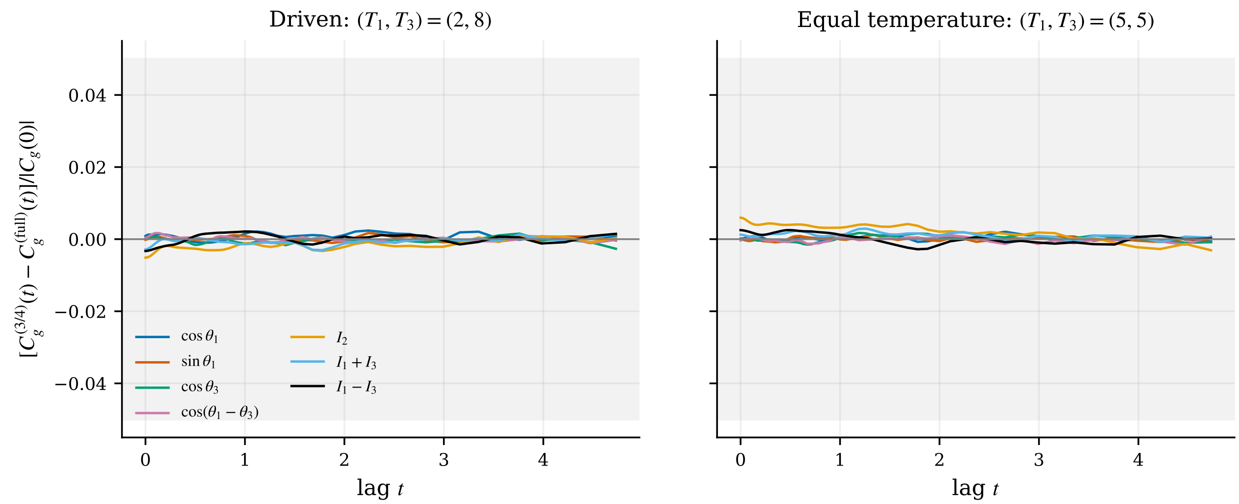

Figure 2: Burn-in check. Differences between full-measurement correlations and correlations after deleting the first quarter, normalized by $|C_g(0)|$. The grey region is the predeclared $\pm5\%$ stationarity tolerance.

Damped-oscillation check

We fitted

$$C(t) = e^{\lambda_R t}\{A\cos(\lambda_I t) + B\sin(\lambda_I t)\}$$

on the predeclared late support. No observable has a resolved nonzero λ_I: every candidate either fails the half-cycle resolution condition, fails to improve AIC by 10 over the nested real exponential, or lacks enough supported points. In particular, the driven $\sin\theta_1$ unconstrained optimum is $\lambda_I = 7.02$ with bootstrap interval approximately $[0, 10.81]$, but the damped model has $\Delta\mathrm{AIC} = +1.43$ and is not accepted.

case	g	fitted λ_I	95% CI	half-cycle threshold	verdict
driven	$\cos\theta_1$	1.3×10^{-10}	$[2.8\times10^{-11}, 2.1\times10^{-5}]$	1.138	not resolved
driven	$\sin\theta_1$	7.019	$[3.0\times10^{-6}, 10.815]$	1.054	AIC fails
driven	$\cos\theta_3$	4.0×10^{-10}	$[4.3\times10^{-11}, 0.0133]$	1.579	not resolved
driven	$\cos(\theta_1-\theta_3)$	8.0×10^{-7}	$[1.8\times10^{-10}, 0.0037]$	1.893	not resolved
driven	I_2	7.8×10^{-8}	$[3.9\times10^{-10}, 0.0055]$	1.054	not resolved
driven	I_1+I_3	7.3×10^{-9}	$[3.9\times10^{-11}, 0.0049]$	1.054	not resolved
driven	I_1-I_3	—	—	—	insufficient support
equilibrium	$\cos\theta_1$	1.6×10^{-10}	$[1.3\times10^{-12}, 0.0064]$	1.454	not resolved
equilibrium	$\sin\theta_1$	—	—	—	insufficient support
equilibrium	$\cos\theta_3$	0.0015	$[8.9\times10^{-11}, 0.0073]$	1.579	not resolved
equilibrium	$\cos(\theta_1-\theta_3)$	7.6×10^{-9}	$[2.0\times10^{-10}, 0.0015]$	1.454	not resolved
equilibrium	I_2	0.00019	$[5.6\times10^{-11}, 0.00040]$	0.905	not resolved
equilibrium	I_1+I_3	3.4×10^{-7}	$[1.1\times10^{-9}, 3.4\times10^{-5}]$	0.680	not resolved
equilibrium	I_1-I_3	—	—	—	insufficient support

Table 2: Damped-fit frequencies. A frequency is accepted only if the interval contains at least half a cycle and the damped model improves AIC by at least 10 over the nested real exponential. None passes.

The machine-readable values, including λ_R, bootstrap acceptance and ΔAIC, are in `analysis/*_damped_fits.csv`. This supports an essentially real resolved late decay, but not a unique real gap.

Comparison with the previous numbers

The two driven plateaus, -0.993 and -1.109, are closest to the historical full-window diagnostic -0.934. They do not reproduce the early-window and Prony values -1.60 and -1.65, and the data do not support interpreting -0.68 as a common late-time plateau. The correct replacement sentence is:

> Stationary autocorrelations resolve observable-specific rates of order -1, but no common late-time plateau across observables; the spectral gap remains numerically unresolved.

Numerical caveat and provenance

No trajectory became non-finite. The inherited floor was nevertheless activated 91,674 times in the driven case and 97,047 times at equilibrium, so this experiment estimates the spectrum of the stated discretized dynamics and does not remove that historical numerical caveat.

The frozen source commit is `b238c2fff9e5f9366cb2c8bafddcef90f7da5b12`; its remote cherry-pick is `db022529c1d8e4e4d1153742018a088d8be3264b`. Sampler source SHA-256 is `a1d558c267a4eca00c3b8f445e1236c1d097be50e72845f18ed4a4e12d7c9df1`; binary SHA-256 is `447ba1141ba5f126bccd6f39661c939eee4683ef96fcd12a1595420e7e532316`. Base seeds are 2026090701 and 2026090702, with stream seed equal to base plus stream ID. Exact commands and complete hashes are under `provenance/`; the raw correlation tables are `analysis/*_autocorrelations.csv`.